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Studierichting: Informatica
Oefeningenreeks 4A nr. 55.3

$$
\begin{aligned}
&\int \tan^5 x dx \\
&= \frac{1}{4}\tan^4 x - \int \tan^3 x dx \\
&= \frac{1}{4}\tan^4 x - \left(\frac{1}{2}\tan^2 x - \int \tan x dx\right) \\
&= \frac{1}{4}\tan^4 x - \left(\frac{1}{2}\tan^2 x - \int \frac{\sin x}{\cos x} dx\right) \\
&\text{Stel } \cos x = t \Rightarrow dt = -\sin x dx \Rightarrow dx = \frac{dt}{-\sin x} \\
&\Rightarrow \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + \int \frac{dt}{-t} = \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \int \frac{dt}{t} \\
&= \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \ln|t| + c \\
&= \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \ln|\cos x| + c
\end{aligned}
$$

References